

# InstantLink / Provisioning & Activation Deployment Guide (vLab, PostgreSQL)

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## Environment Details

- **Client Tool:** MobaXterm
  - **Server:** nipo-test-1
  - **OS:** baseos-23.10.2-785.os8.x86\_64
  - **IP:** 100.84.119.60
  - **vLab Session:** c10\_r24\_d250
  - **Lab Name:** nippo-lab
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## Step 0: Pre-Requisites (Run as root)

**Purpose:** Ensure OS has all required packages for PostgreSQL and InstantLink.

```
# Set proxy if required
export http_proxy=http://87.254.212.120:8080
export https_proxy=https://87.254.212.120:8080
```

```
# Install essential tools and libraries
dnf install -y bc binutils fontconfig-devel glibc-devel ksh
libaio \
libaio-devel libgcc libstdc++ libstdc++-devel make sysstat unzip
wget \
postgresql-server postgresql-contrib --skip-broken
```

**Outcome:** Environment ready with KornShell, PostgreSQL packages, and essential libraries.

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## Step 1: Create Dedicated User & Group for InstantLink (Run as root)

```
groupadd telecom
mkdir -p /telecom/ilink
chown root:telecom /telecom
chmod 755 /telecom
```

```
useradd -g telecom -d /telecom/ilink -s /bin/ksh -M ilink
passwd ilink
chown -R ilink:telecom /telecom/ilink
id ilink
```

**Outcome:** ilink user ready with isolated home directory.

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## Step 2: PostgreSQL Database Setup (Run as root)

**Note:** PostgreSQL RPM packages are installed locally and uploaded to vLab.

### Initialize and Start PostgreSQL

```
postgresql-setup --initdb
systemctl enable postgresql
systemctl start postgresql
systemctl status postgresql
```

### Create Database and User for InstantLink

```
sudo -i -u postgres
psql
DROP DATABASE IF EXISTS ilink_db;
DROP USER IF EXISTS ilink_user;
CREATE DATABASE ilink_db;
CREATE USER ilink_user WITH PASSWORD 'IlinkPass0465';
GRANT ALL PRIVILEGES ON DATABASE ilink_db TO ilink_user;
\q
exit
```

### Set PostgreSQL Environment Variables (Run as ilink)

```
su - ilink
export PGHOST=localhost
export PGPORT=5432
export PGUSER=ilink_user
export PGPASSWORD=IlinkPass0465
export PGDATABASE=ilink_db

# Make permanent
echo "export PGHOST=localhost" >> /telecom/ilink/.profile
echo "export PGPORT=5432" >> /telecom/ilink/.profile
echo "export PGUSER=ilink_user" >> /telecom/ilink/.profile
echo "export PGPASSWORD=IlinkPass0465" >> /telecom/ilink/.profile
echo "export PGDATABASE=ilink_db" >> /telecom/ilink/.profile
```

**Outcome:** ilink user can connect to PostgreSQL automatically.

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### Step 3: Prepare Directories (Run as root)

```
mkdir -p /telecom/release_area
chown -R ilink:telecom /telecom/release_area
chmod 755 /telecom/release_area
```

**Outcome:** Staging area ready for packages.

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### Step 4: Deploy InstantLink / P&A (Run as ilink)

```
su - ilink
cd /telecom/release_area
# Upload local packages to vLab and extract
cp /local/path/INSTANTLINK-25.0.0_545.cpio.Z /telecom/release_area/
cpio -idmv < INSTANTLINK-25.0.0_545.cpio.Z
```

#### Configure pa-solution.config

```
host_name=
jre_home=/usr/lib/jvm/java-17-openjdk
pa_db_type=postgresql
pa_db_home=/var/lib/pgsql
# Remove jdbc path from URL as it works without it
pa_db_url=postgresql://localhost:5432/ilink_db
pa_db_user=ilink_user
pa_db_pass=IlinkPass0465
il_http_port=44080
il_https_port=44043
```

#### Run Installer

```
./manage_releases -s ./pa-solution.config -i /telecom/ilink -r
/telecom/release_area
```

#### Optional Cleanup Before Reinstall:

```
rm -rf /telecom/ilink/* /telecom/release_area/*
cp /telecom/packages/installation_specifications/pa-solution.config
/telecom/packages/installation_specifications/pa-solution.config.bak
vi /telecom/packages/installation_specifications/pa-solution.config
```

**Outcome:** InstantLink installed in /telecom/ilink.

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## Step 5: Verify Installation (Run as ilink)

```
. /telecom/ilink/sas/.profile
ctl_control -s ALL start
ctl_control -s ALL status
psql -h localhost -U ilink_user -d ilink_db -c "SELECT NOW();"
```

**Outcome:** All components running; PostgreSQL DB connected.

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## Step 6: Access InstantLink UI

- Browser: <http://<vLab-IP>:44080>
- Default credentials: Administrator / guiadmin
- If ports blocked, use SSH tunnel:

```
ssh -L 8080:localhost:44080 root@<vLab-IP>
```

Then access: <http://localhost:8080>

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## Summary Table

| Component             | Purpose                                  | Run As                          |
|-----------------------|------------------------------------------|---------------------------------|
| Environment Setup     | Install OS packages, user/group creation | root                            |
| ilink                 | Extract packages, run installer          | ilink                           |
| PostgreSQL DB         | Store provisioning data                  | root (install), ilink (connect) |
| /telecom/release_area | Staging of packages                      | ilink                           |
| /telecom/ilink        | Product home                             | ilink                           |
| Web UI                | Provisioning management                  | Browser                         |

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## Safety Notes

- Commands isolated to your vLab session.
  - **ilink** user never requires root after initial setup.
  - Conflicting OS packages skipped.
  - PostgreSQL runs under its system user, only ilink uses the schema.
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## Troubleshooting

- Check logs: `tail -n 50 /telecom/release_area/runs/run_YYYY-MM-DD/install_output.log`
- Recreate DB if needed:

```
sudo -i -u postgres
```

```
psql
```

```
DROP DATABASE IF EXISTS ilink_db;
```

```
DROP USER IF EXISTS ilink_user;
```

```
CREATE DATABASE ilink_db;
```

```
CREATE USER ilink_user WITH PASSWORD 'IlinkPass0465';
```

```
GRANT ALL PRIVILEGES ON DATABASE ilink_db TO ilink_user;
```

```
\q
```

```
exit
```

- Reload environment if `.profile` changed: `. /telecom/ilink/sas/.profile`
- Restart components: `ctl_control -s ALL start`